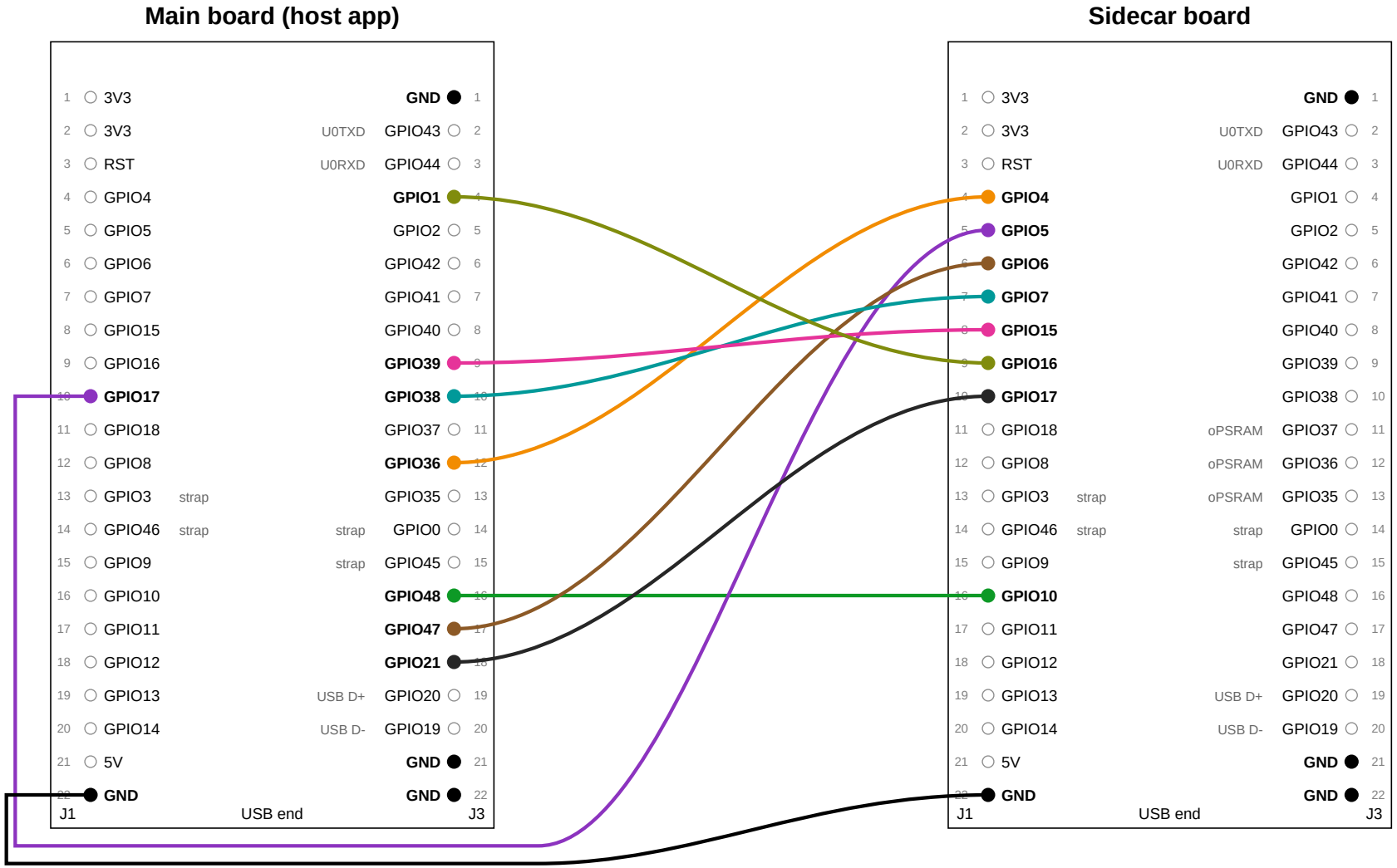
Sidecar bus sampler wiring (bare main board, no link)

Generated 2026-11-12 by hardware/tools/sidecar_wiring.py. Main board: a bare test-pool board (e.g. miner6 .5), Sidecar link off. Header pin numbers: DevKitC-1, pin 1 away from the USB end.

Wire ONLY these 8 signals + GND, 330-470 ohm in series. NO link wires: main GPIO21 is refresh_ here. Sidecar on USB power whenever the Atari is on.



Signal	From	To	Header (from -> to)	Status	Notes
phi2	main GPIO36	Sidecar GPIO4	J3-12 -> J1-4	wire	bit 0: clock
profile	main GPIO48	Sidecar GPIO10	J3-16 -> J1-16	wire	bit 1: main's profile pin (BCLK wire when the link is used)
R/W	main GPIO17	Sidecar GPIO5	J1-10 -> J1-5	wire	bit 2
extSel_	main GPIO47	Sidecar GPIO6	J3-17 -> J1-6	wire	bit 3: open drain on the main board, keep it short
D0	main GPIO38	Sidecar GPIO7	J3-10 -> J1-7	wire	bit 4
D1	main GPIO39	Sidecar GPIO15	J3-9 -> J1-8	wire	bit 5
A0	main GPIO1	Sidecar GPIO16	J3-4 -> J1-9	wire	bit 6
refresh_	main GPIO21	Sidecar GPIO17	J3-18 -> J1-10	wire	bit 7: main GPIO21 is refresh_ on this board

Sidecar bus sampler wiring (bare main board, no link) (continued)

Signal	From	To	Header (from -> to)	Status	Notes
GND	main GND	Sidecar GND	J1-22 -> J1-22	wire	common ground: any GND pin on each board

Solid: wired now. Dashed: planned. Filled pin: has a wire. oPSRAM: octal-PSRAM pin (N8R8, N16R8 modules).

Wire ONLY these nine (8 signals + GND). Do NOT wire the link: main GPIO21 is refresh_ here, and the Sidecar's link TX (GPIO46) would fight the Atari's RFSH. Main GPIO48 is the profile pin with the link off.
Bus-profiler taps: Sidecar inputs only, no pulls; 330-470 ohm in series at the main board end.
Keep the Sidecar powered (its own USB) whenever the Atari is on: an unpowered ESP32 loads the bus lines.